Filbin et al. 2018 — Pediatric H3-K27M Glioma

CCCA meta-atlas · embryonic-program cell-state analysis

GO / NO-GO GATES

Gate 1 **GO** Mixing & continuity — on the θ sweep: θ=0 is a per-patient archipelago; θ=1 fully intermixes all six patients into one connected manifold (cross-sample kNN 0.03→0.61); 0 detached islands.

Gate 2 **GO** Module geography — after scoping out the cycling lobe (Leiden clusters majority cell-cycle-positive, ~13% of cells), the embryonic manifold still branches; Gast & pSom co-vary (smoothed +0.80), Crest is the inverse terminal gradient (high at a differentiated arm tip).

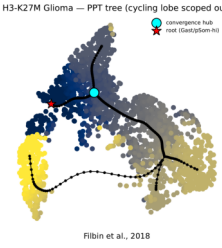
Gate 3 **NO-GO** Cross-repression — the tree branches, but the split is by CREST, not Gast↔pSom. Every arm is pSom-positive (+0.12…+0.25) and Gast-flat (−0.08…+0.05): no Gast-arm vs pSom-arm. Two arms are Crest-terminal. Convergence-hub per-gene Gast↔pSom r = +0.26 (positive, far from Puram −0.49). → NO-GO on step 05.

STEP FIGURES

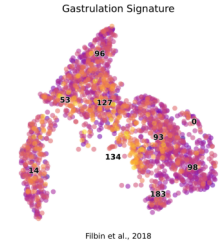
Samples (Gate 1 · integration)



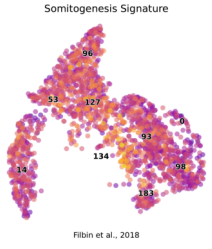
Manifold — tree, cycling lobe scoped



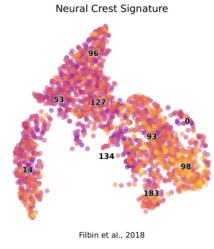
Gast signature



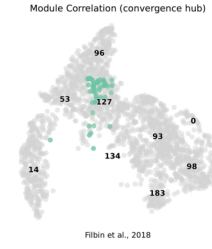
pSom signature



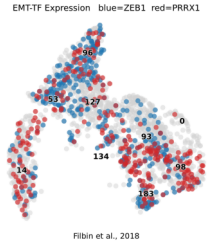
Crest signature



Hub module correlation (Gate 3)



EMT-TF expression



Dataset. Pediatric diffuse midline glioma (H3-K27M) · SmartSeq2 (TPM) · 2,259 malignant cells, 6 patients.

Verdict. Branching embryonic manifold (OPC/OC/AC hierarchy) after scoping out the cycling lobe, but NO Gast↔pSom cross-repression at the convergence hub — the tree separates Crest-terminal states, not competing embryonic arms.

Parameter choices

Min-cell threshold	none (no clean count gap; smallest sample 141 > ~100 floor, integrates)
Scoping	cycling lobe (majority-cycling Leiden clusters, ~13%) removed BEFORE fitting -> 1,967-cell embryonic manifold
Geometry	Branching (scFates PPT tree on scoped X_pca_harmony; Nodes=200 λ=2000 σ=0.5; 5 tips, 3 forks)
Embedding	Harmony θ=1 on sample · n_pcs=18 · native X_umap for viewing (no re-embedding)
Root / conv. hub	root = Gast/pSom-high Crest-low tip (node 53); convergence hub = max-betweenness fork (node 127)
Annotation	Filbin OPC/OC/AC-like states (published) + lineage markers (data-derived)
Hub r (Gate 3)	+0.26 (per-gene, convergence-hub geodesic nbhd) — positive, no cross-repression
EMT-TFs shown	ZEB1, PRRX1 (most-separating detectable; descriptive, no order asserted)